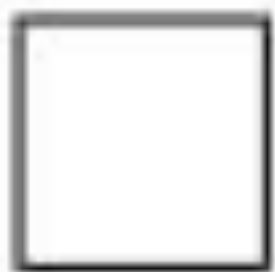


Chapter 24

Organic and biological chemistry

1. If each of the following represents a cycloalkanes, and a carbon atom is located at each vertex with the proper number of hydrogen atoms also bonded to it, which one is the most reactive? D

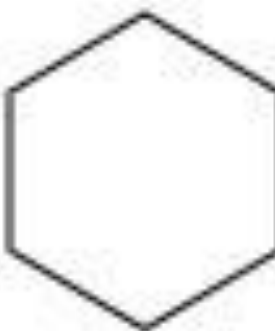
A)



B)



C)



D)



2. How many isomers are possible for C_5H_{12} ? C

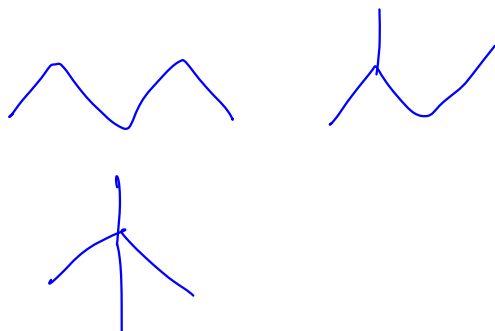
A) 1

B) 2

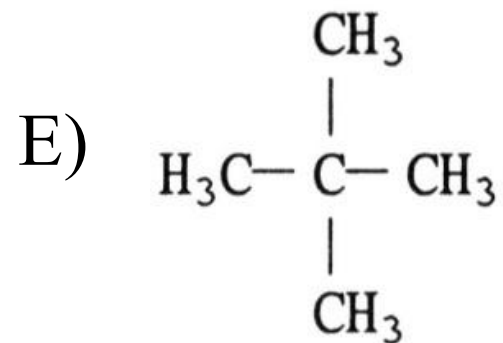
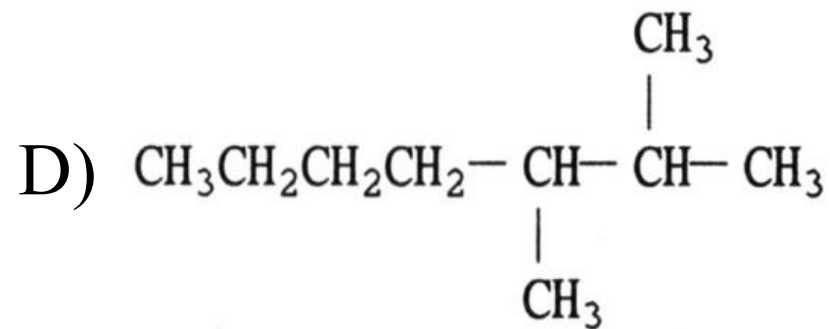
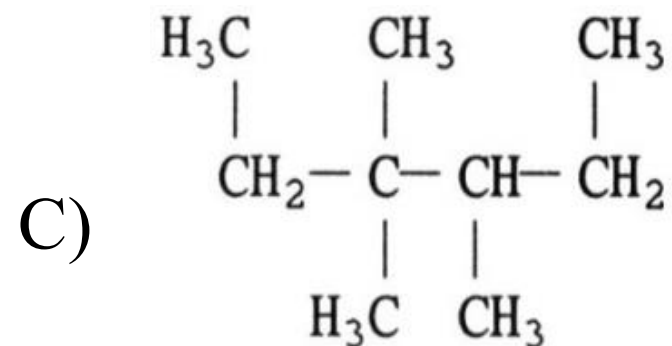
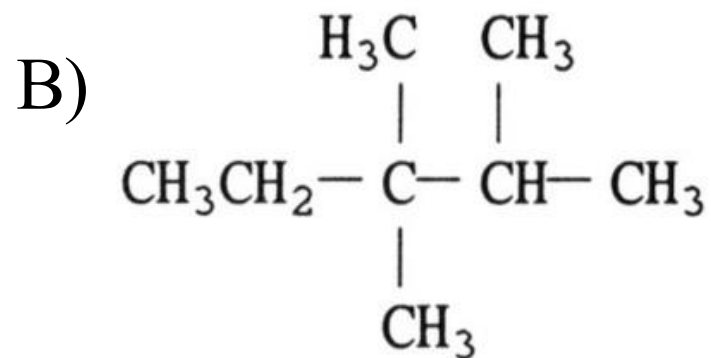
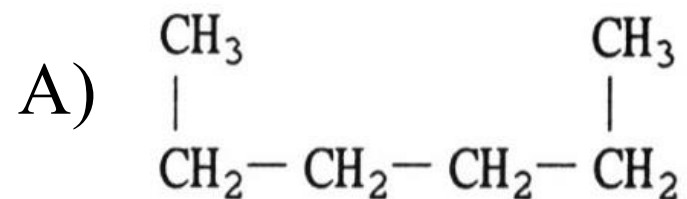
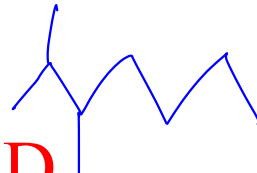
C) 3

D) 4

E) 10



3. The structure of 2,3-dimethylheptane is D.



4. Which statement about hydrocarbons is false? C

A) The smallest alkane to have structural (constitutional) isomers has 4 carbon atoms. ✓

B) Cyclic alkanes are structural isomers of alkenes. ✓

C) Alkanes are more reactive than ^{Double bonds} alkenes. ✗

D) Alkanes can be produced by hydrogenating alkenes. ✓

E) Alkenes can be polymerized. ✓

5. Benzene behaves differently from a hydrocarbon which simply contains three C=C bonds in that the latter would be expected to react much more readily with E.

A) H₂

B) Cl₂

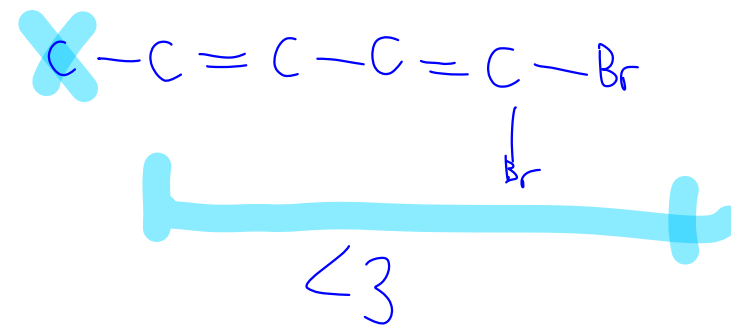
C) Br₂

D) HCl

E) all of the above

6. How many chiral centers are there in $\text{CH}_3\text{CHCHCH}_2\text{CHBr}_2$? A

- A) 0
- B) 1
- C) 2
- D) 3
- E) 4



7. Sugars are examples of what type of molecule? **B**

A) proteins

B) carbohydrates

C) nucleic acids

D) amino acids

E) salts

8. The double helix of DNA is stabilized mainly by E.

A) ionic bonds

B) covalent bonds

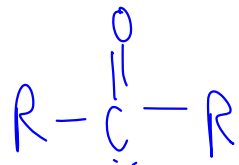
C) ion-dipole bonds

D) ester bonds

E) hydrogen bonds

9. Which of the following compounds do not contain an sp^3 hybridized oxygen atom? A

A) ketones



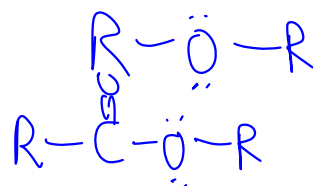
B) alcohols



C) ethers



D) esters

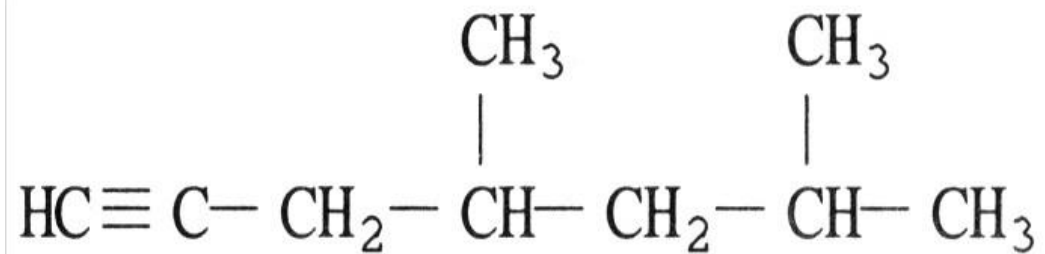


E) water



10. Give IUPAC names for the following compounds.

a.



Answer:

a. 4,6-dimethyl-1-heptyne

b.



b. 2,5-octadiene